

BUSINESS CASE: NETFLIX - DATA EXPLORATION AND VISUALISATION

The objective of this case study is to analyze Netflix's content dataset to understand trends in content type, Genres, Countries and release patterns in order to recommend what type of content Netflix should produce and where to focus for business growth.

Load and Understand the Data:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Read csv File:

```
data=pd.read_csv("netflix.csv")
data
```

show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description	
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town L...
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train L...
...	...	...	...	...	...	...	...	...	...	...	...	
8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	November 20, 2019	2007	R	158 min	Cult Movies, Dramas, Thrillers	A political cartoonist, a crime reporter and a...
8803	s8804	TV Show	Zombie Dumb	NaN	NaN	NaN	July 1, 2019	2018	TV-Y7	2 Seasons	Kids' TV, Korean TV Shows, TV Comedies	While living alone in a spooky town, a young g...
8804	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	November 1, 2019	2009	R	88 min	Comedies, Horror Movies	Looking to survive in a world taken over by zo...
8805	s8806	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	January 11, 2020	2006	PG	88 min	Children & Family Movies, Comedies	Dragged from civilian life, a former superhero...
8806	s8807	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	March 2, 2019	2015	TV-14	111 min	Dramas, International Movies, Music & Musicals	A scrappy but poor boy worms his way into a ty...
8807 rows × 12 columns												

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Top 5 data:

```
data.head()
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mablane, Thaban...	South Africa	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town l...
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train l...

Data Columns:

```
data.columns
```

```
*** Index(['show_id', 'type', 'title', 'director', 'cast', 'country', 'date_added',  
          'release_year', 'rating', 'duration', 'listed_in', 'description'],  
          dtype='object')
```

Data Shape:

```
data.shape
```

```
*** (8807, 12)
```

Data Dimension:

```
data.ndim
```

```
2
```

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Data Information:

```
data.info()
```

```
*** <class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  ---                ---
0   show_id               8807 non-null   object
1   type                 8807 non-null   object
2   title                8807 non-null   object
3   director             6173 non-null   object
4   cast                 7982 non-null   object
5   country              7976 non-null   object
6   date_added           8797 non-null   object
7   release_year         8807 non-null   int64
8   rating               8803 non-null   object
9   duration             8804 non-null   object
10  listed_in            8807 non-null   object
11  description           8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

Checking for Null values:

```
data.isnull().sum()
```

```
***
```

	0
show_id	0
type	0
title	0
director	2634
cast	825
country	831
date_added	10
release_year	0
rating	4
duration	3
listed_in	0
description	0

dtype: int64

Since from the above table we can see that there are some null values present so we will convert that null values into some meaningful value and null values which are not important we drop them out

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Converting date_added to datetime:

```
data["date_added"] = pd.to_datetime(data["date_added"],  
errors="coerce")
```

Dropping rows where date or duration is missing:

```
data = data.dropna(subset=["date_added", "duration"])
```

"Invalid or missing values in the date_added column were converted to NaT(Not a Time)and removed to ensure reliable time-based analysis".

Filling other categorical columns:

```
data.loc[:, 'director'] = data["director"].fillna("Unknown  
Director")  
data.loc[:, 'cast'] = data["cast"].fillna("Unknown Actor")  
data.loc[:, 'country'] = data["country"].fillna("Unknown  
Country")  
data.loc[:, 'rating'] = data["rating"].fillna("Not-Rated")
```

"Here in a categorical column, we use Unknown Director, Unknown Actor and Unknown Country and for rating column we give Not-Rated for a null value".

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Checking Null values:

```
data.isnull().sum()
```

...

	index	Count
0	show_id	0
1	type	0
2	title	0
3	director	0
4	cast	0
5	country	0
6	date_added	0
7	release_year	0
8	rating	0
9	duration	0
10	listed_in	0
11	description	0

As we can see that the sum of null values as been zero so now we can say that the missing values in the categorical identity fields (director, cast, country) were replaced with "Unknown Director", "Unknown Actor" and "Unknown Country". Missing ratings were replaced with "Not-Rated". Rows with missing duration or date_added were removed due to their importance in temporal and content-length analysis.

Extracting Year from date added column:

```
data.loc[:, "Year_added"] = data["date_added"].dt.year  
data.loc[:, "Month_added"] = data["date_added"].dt.month
```

We created a derived feature "Year_added" from date_added to perform temporal analysis. This allows aggregation of content additions by year, helping identify platform growth trends, content strategy shifts and data-driven recommendations for business decision.

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New Netflix Data Check:

data

...	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description	Year_added	Month_added
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	Unknown Actor	United States	2021-09-25	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...	2021	9
1	s2	TV Show	Blood & Water	Unknown Director	Ama Oamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...	2021	9
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	Unknown Country	2021-09-24	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...	2021	9
3	s4	TV Show	Jailbirds New Orleans	Unknown Director	Unknown Actor	Unknown Country	2021-09-24	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...	2021	9
4	s5	TV Show	Kota Factory	Unknown Director	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train I...	2021	9
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	2019-11-20	2007	R	158 min	Cult Movies, Dramas, Thrillers	A political cartoonist, a crime reporter and a...	2019	11
8803	s8804	TV Show	Zombie Dumb	Unknown Director	Unknown Actor	Unknown Country	2019-07-01	2018	TV-Y7	2 Seasons	Kids' TV, Korean TV Shows, TV Comedies	While living alone in a spooky town, a young g...	2019	7
8804	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	2019-11-01	2009	R	88 min	Comedies, Horror Movies	Looking to survive in a world taken over by zo...	2019	11
8805	s8806	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	2020-01-11	2006	PG	88 min	Children & Family Movies, Comedies	Dragged from civilian life, a former superhero...	2020	1
8806	s8807	Movie	Zubaan	Moze Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	2019-03-02	2015	TV-14	111 min	Dramas, International Movies, Music & Musicals	A scrappy but poor boy worms his way into a ty...	2019	3

8706 rows x 14 columns

Here we can see that after cleaning up the data we can see that where ever we did a modification is been done and now we are ready for the analysis of the data.

Counts of Type:

```
data["type"].value_counts().reset_index()
```

...	type	count
0	Movie	6128
1	TV Show	2578

Insights

The dataset contains two contents types:
Movies and TV Shows.

- Movies (6128) constitute the majority of content available on the platform.
- TV Shows (2578) represent a smaller but significant portion of the catalog.

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- The higher volume of movies suggests a stronger focus on film acquisition and production compared to serialized content.

Business Implication: The platform appears to follow a movie-dominant content strategy, which may be driven by broader audience appeal and lower production complexity compared to long-format TV series.

Counts of Rating:

```
data["rating"].value_counts().reset_index()
```

...	rating	count
0	TV-MA	3183
1	TV-14	2133
2	TV-PG	838
3	R	799
4	PG-13	490
5	TV-Y7	330
6	TV-Y	300
7	PG	287
8	TV-G	212
9	NR	78
10	G	41
11	TV-Y7-FV	5
12	Not-Rated	4
13	NC-17	3
14	UR	3

Insights

- TV-MA (Mature Audience) is the most frequent content rating .
- NC-17(No Children Under 17 Admitted) and UR(Unrated) appear very rarely.
- The distribution indicates a strong focus on mature and young-adult audiences.

Business Implication: Netflix's primarily targets adult viewers, aligning content strategy with global viewing preferences.

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Number of Unique Countries:

```
data["country"].nunique()  
...    746
```

Insights

- There are 195 countries globally, while the datasets shows 746 country entries.
- This count exceeds the total number of countries because many titles are associated with multiple countries due to international co-productions.
- This reflects Netflix's strong focus on cross-border collaborations and globally distributed content.

Business Implication: Netflix leverages international partnerships to expand content reach, localize offerings and appeal to diverse audiences across multiple regions.

Description of Release Year:

```
data["release_year"].describe()  
...  
      release_year  
count    8706.000000  
mean     2014.197105  
std       8.827570  
min       1925.000000  
25%      2013.000000  
50%      2017.000000  
75%      2019.000000  
max       2021.000000  
  
dtype: float64
```

Insights

- The dataset contains 8,706 titles with release years ranging from 1925 to 2021.
- The average release year is approximately 2014, indicating that most content is relatively recent.
- 75% of the titles were release after 2013 and the median release year is 2017.
- The standard deviation of 8.8years suggests moderate concentration around recent years.

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Business Implication: Netflix's catalog is heavily skewed toward modern content with majority of titles produced in the last decade. This indicates a strategic focus on acquiring and promoting contemporary content rather than older archival production.

Number of Unique Genre Combination:

```
data["listed_in"].nunique()
```

```
*** 509
```

Insights

- The dataset includes **509 unique Genre Combination** under the listed_in column.
- This indicates a broad variety of content types such as drama, comedies, documentaries, thrillers and international genre.
- The large number of genre combinations shows Netflix's emphasis on content diversity rather than focusing on a few categories.
- It allows the platform to cater to varied viewer preference and viewing mood.

Business Implication: A diverse genre portfolio helps Netflix attract a wider audience base and improve user retention by offering content across multiple interest.

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Country-wise Content:

```
data_country = data.assign(country =  
data["country"].str.split(",").explode("country"))
```

data_country

...	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description	Year_added	Month_added	
	0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	Unknown Actor	United States	2021-09-25	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...	2021	9
	1	s2	TV Show	Blood & Water	Unknown Director	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...	2021	9
	2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	Unknown Country	2021-09-24	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...	2021	9
	3	s4	TV Show	Jailbirds New Orleans	Unknown Director	Unknown Actor	Unknown Country	2021-09-24	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...	2021	9
	4	s5	TV Show	Kota Factory	Unknown Director	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train I...	2021	9
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
	8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	2019-11-20	2007	R	158 min	Cult Movies, Dramas, Thrillers	A political cartoonist, a crime reporter and a...	2019	11
	8803	s8804	TV Show	Zombie Dumb	Unknown Director	Unknown Actor	Unknown Country	2019-07-01	2018	TV-Y7	2 Seasons	Kids' TV, Korean TV Shows, TV Comedies	While living alone in a spooky town, a young g...	2019	7
	8804	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	2019-11-01	2009	R	88 min	Comedies, Horror Movies	Looking to survive in a world taken over by zo...	2019	11
	8805	s8806	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	2020-01-11	2006	PG	88 min	Children & Family Movies, Comedies	Dragged from civilian life, a former superhero...	2020	1
	8806	s8807	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah- Jane Dias, Raaghav Chanan...	India	2019-03-02	2015	TV-14	111 min	Dramas, International Movies, Music & Musicals	A scrappy but poor boy worms his way into a ty...	2019	3
10736 rows x 14 columns															

Top 10 Countries:

```
top_10_countries =  
(data_country["country"].value_counts().head(10))  
top_10_countries.reset_index()
```

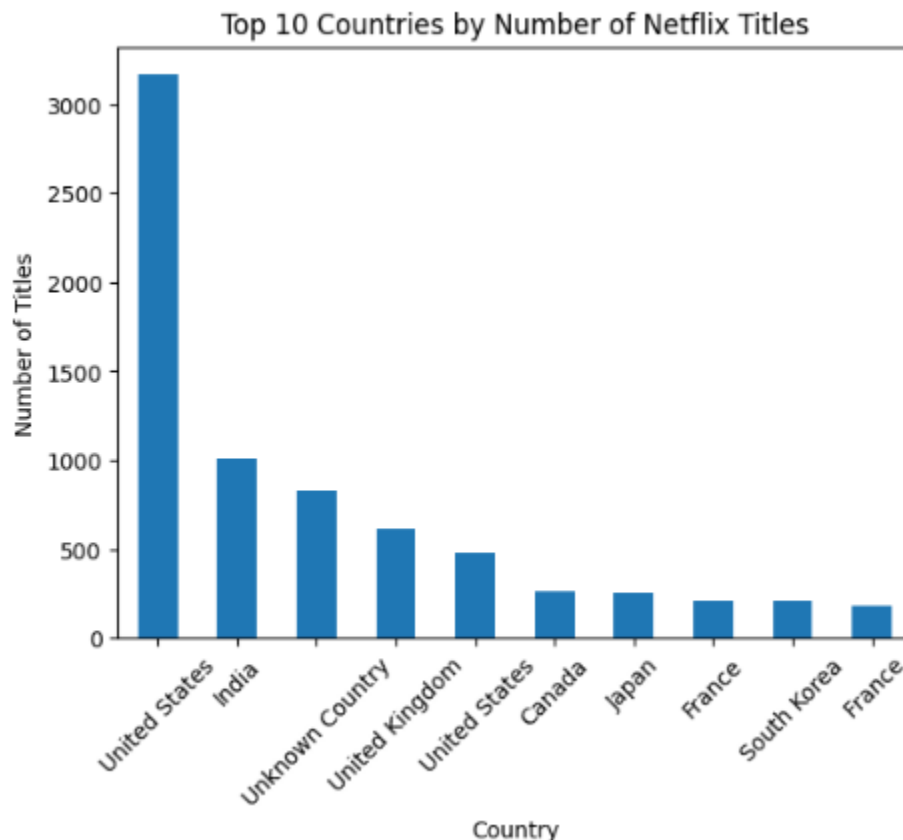
...	country	count
0	United States	3166
1	India	1007
2	Unknown Country	827
3	United Kingdom	611
4	United States	474
5	Canada	259
6	Japan	255
7	France	209
8	South Korea	206
9	France	180

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Top 10 Countries by number of Netflix titles:

```
top_10_countries.plot(kind="bar")
plt.xlabel("Country")
plt.ylabel("Number of Titles")
plt.xticks(rotation=45)
plt.title("Top 10 Countries by Number of Netflix Titles")
plt.show()
```

...



Insights

- **United States** has the **highest number of titles (approx. 3166)**, contributing **by far the largest share** of Netflix content.
- **India** ranks **second (approx. 1007 titles)**, indicating Netflix's strong focus on Indian market.
- **"Unknown Country" (approx. 827 titles)** appears prominently, suggesting **incomplete or missing country metadata** for a significant portion of content.
- Among non-US market, **United Kingdom (approx. 611)**, **Canada (approx. 259)**, **Japan (255)**, **France (209)** and **South Korea (206)** are key contributors.
- There is a **steep drop** after the top 2 countries, indicating that Netflix's content production is **highly concentrated in a few regions**.

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Business Implication: Netflix's content library is heavily dominated by the United States which alone accounts for a substantial portion of total titles. India emerges as the second-largest contributor, highlighting Netflix's strategic investment in Indian Market. The presence of a large number titles under "Unknown Country" indicates gaps in metadata quality, which may impact region-based analysis and recommendation. Apart from US and India, countries such as UK, Canada, Japan, France and South Korea contribute moderately suggesting selective but growing regional diversification. Overall, Netflix's content distribution is concentrated in a few major markets leaving scope for expansion in under-represented regions.

Number of Titles added over Years:

```
Yearly_titles =  
data["Year_added"].value_counts().sort_index().reset_index()
```

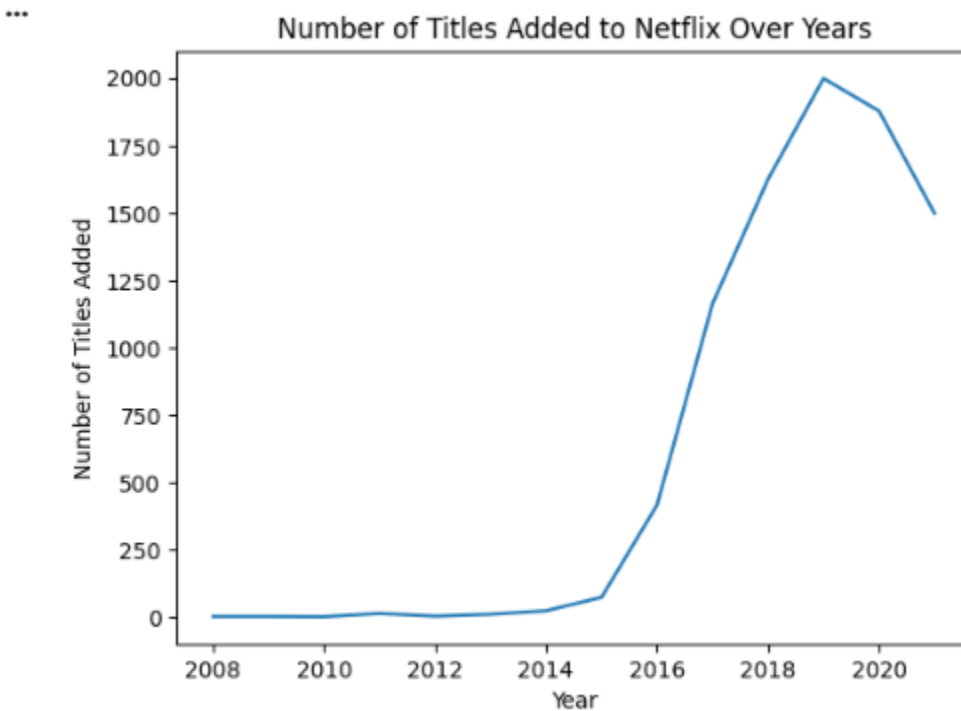
Yearly_titles

***	Year_added	count	
0	2008	2	
1	2009	2	
2	2010	1	
3	2011	13	
4	2012	3	
5	2013	10	
6	2014	23	
7	2015	73	
8	2016	416	
9	2017	1163	
10	2018	1625	
11	2019	1999	
12	2020	1878	
13	2021	1498	

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Plot Titles added on Netflix over the year:

```
sns.lineplot(data = Yearly_titles, x="Year_added", y="count")
plt.xlabel("Year")
plt.ylabel("Number of Titles Added")
plt.title("Number of Titles Added to Netflix Over Years")
plt.show()
```



Insights

- Netflix added relatively few titles before **2015**, indicating a smaller content library during its early years.
- From **2016 onwards**, there is a **sharp and sustained increase** in content additions, reflecting aggressive expansion and investments in original and licensed content.
- The **peak year** for content addition is **2019**, with approximately **2,000 titles added**, marking Netflix's most aggressive growth phase.
- After **2020**, the number of titles added **declines**, suggesting a shift in strategy or external constraints on production.

Business Implication:

- The rapid growth after 2016 indicates that **content volume played a critical role in user acquisition and market penetration**, especially in international markets.

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- The peak in 2019 suggest Netflix may have **reached a saturation point**, after which adding more content yielded diminishing returns.
- The post-2020 decline implies a **strategic transition from quantity-driven growth to quality-focused curation**, aiming to improve user engagement and retention rather than sheer catalog size.
- Going forward, Netflix should focus on **high-performing genres and regions** rather than increasing total volume, Optimize content budgets by investing in **data-backed originals** instead of mass acquisitions and align content release strategy with **viewer demand cycles** rather than aggressive yearly expansion.

Movies vs TV Shows:

```
Movie_Tv_Trend =  
(data.groupby(["Year_added", "type"]).size().reset_index(name =  
"count"))
```

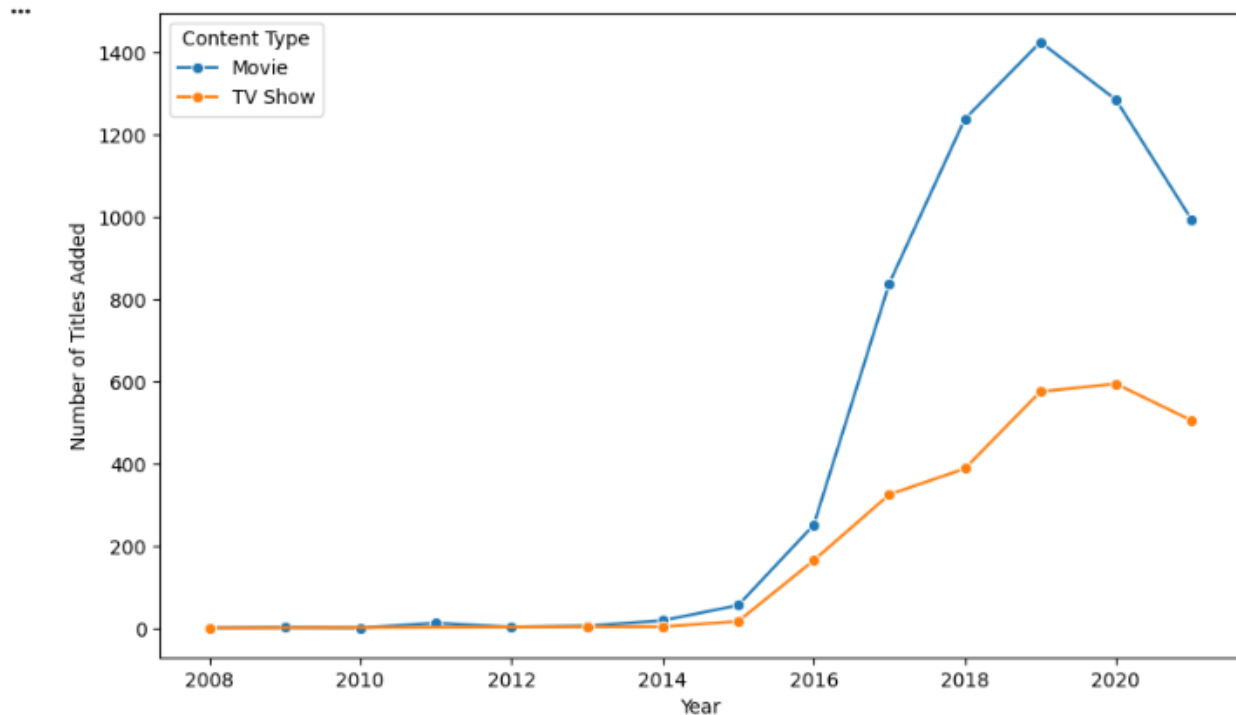
Movie_Tv_Trend

	Year_added	type	count
0	2008	Movie	1
1	2008	TV Show	1
2	2009	Movie	2
3	2010	Movie	1
4	2011	Movie	13
5	2012	Movie	3
6	2013	Movie	6
7	2013	TV Show	4
8	2014	Movie	19
9	2014	TV Show	4
10	2015	Movie	56
11	2015	TV Show	17
12	2016	Movie	251
13	2016	TV Show	165
14	2017	Movie	838
15	2017	TV Show	325
16	2018	Movie	1237
17	2018	TV Show	388
18	2019	Movie	1424
19	2019	TV Show	575
20	2020	Movie	1284
21	2020	TV Show	594
22	2021	Movie	993
23	2021	TV Show	505

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Plot Trend of Content Type:

```
plt.figure(figsize = (10,6))
sns.lineplot(data = Movie_Tv_Trend, x = "Year_added", y =
"count", hue = "type", marker = "o")
plt.xlabel("Year")
plt.ylabel("Number of Titles Added")
plt.legend(title = "Content Type")
plt.show()
```



Insights

Netflix's content expansion accelerated significantly after 2016, with Movies consistently forming the majority of titles. However, TV Shows show strong growth trajectory, indicating a strategic shift toward episodic content that supports long-term user engagement. The decline in content additions after 2020 suggests a transition from rapid expansion to a more selective content strategy.

Business Implication:

Netflix should continue investing in TV Shows to strengthen subscriber retention and binge-watch behavior, while using Movies selectively for regional reach and brand visibility. The recent slowdown highlights the need to prioritize content quality and performance over sheer volume.

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Trend Over Years:

```
Yearly_titles = (data["Year_added"].value_counts().sort_index())
```

```
Yearly_titles.reset_index()
```

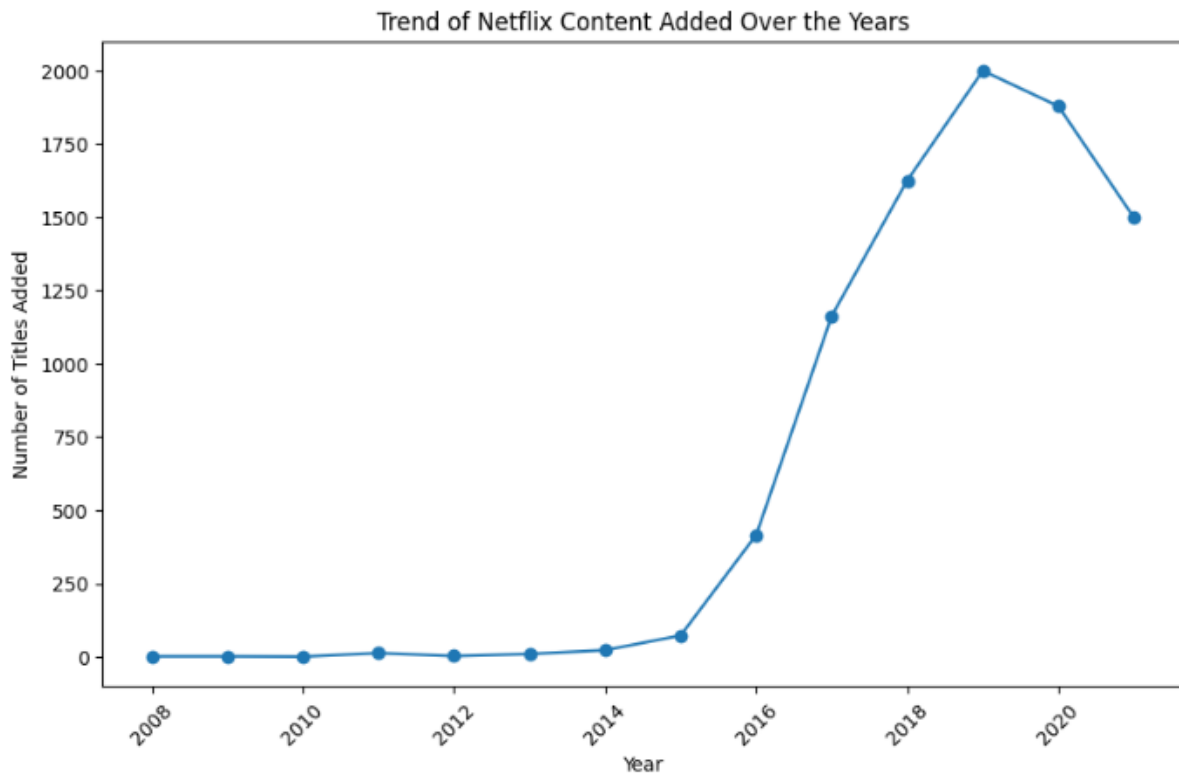
	Year_added	count
0	2008	2
1	2009	2
2	2010	1
3	2011	13
4	2012	3
5	2013	10
6	2014	23
7	2015	73
8	2016	416
9	2017	1163
10	2018	1625
11	2019	1999
12	2020	1878
13	2021	1498

Plot Trend of Netflix Content:

```
plt.figure(figsize=(10,6))
Yearly_titles.plot(kind = "line", marker = "o")
plt.xlabel("Year")
plt.ylabel("Number of Titles Added")
plt.title("Trend of Netflix Content Added Over the Years")
plt.xticks(rotation=45)
plt.show()
```


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...



Insights

Netflix's content additions remained very limited until around **2014**, indicating a conservative content acquisition strategy in its early years. From **2015 onwards**, there is a **sharp and sustained increase**, with the platform experiencing rapid expansion between **2016 and 2019**. The number of titles added **peaks around 2019**, followed by a noticeable **decline in 2020-2021**.

This trend reflects Netflix's transition from a niche streaming service to a global content powerhouse, followed by strategic slowdown in content additions.

Business Implication:

- The rapid growth phase (2016-2019) suggests aggressive market expansion and global subscriber acquisition.
- The decline after 2019 indicates a shift from **quality-driven growth** to **quality-focused content strategy**, possibly influenced by: Market saturation in key regions, Rising content production costs or by COVID-19 related production disruption.
- Going forward, Netflix should focus on **high-impact content investments**, prioritizing regional originals and franchises that drive long-term user retention rather than increasing content volume indiscriminately.

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Month-Wise count of titles:

```
Month_trend =  
(data["Month_added"].value_counts().sort_index().reset_index())  
Month_trend.columns = ["Month", "Count"]
```

Month_trend

***	Month	Count
0	1	727
1	2	557
2	3	734
3	4	758
4	5	626
5	6	724
6	7	819
7	8	748
8	9	764
9	10	755
10	11	697
11	12	797

Converting Month Number into Month Name:

```
Month_map = {1: "January", 2: "February", 3: "March", 4:  
"April", 5: "May", 6: "June", 7: "July", 8: "August",  
9: "September", 10: "October", 11: "November", 12:  
"December"}
```

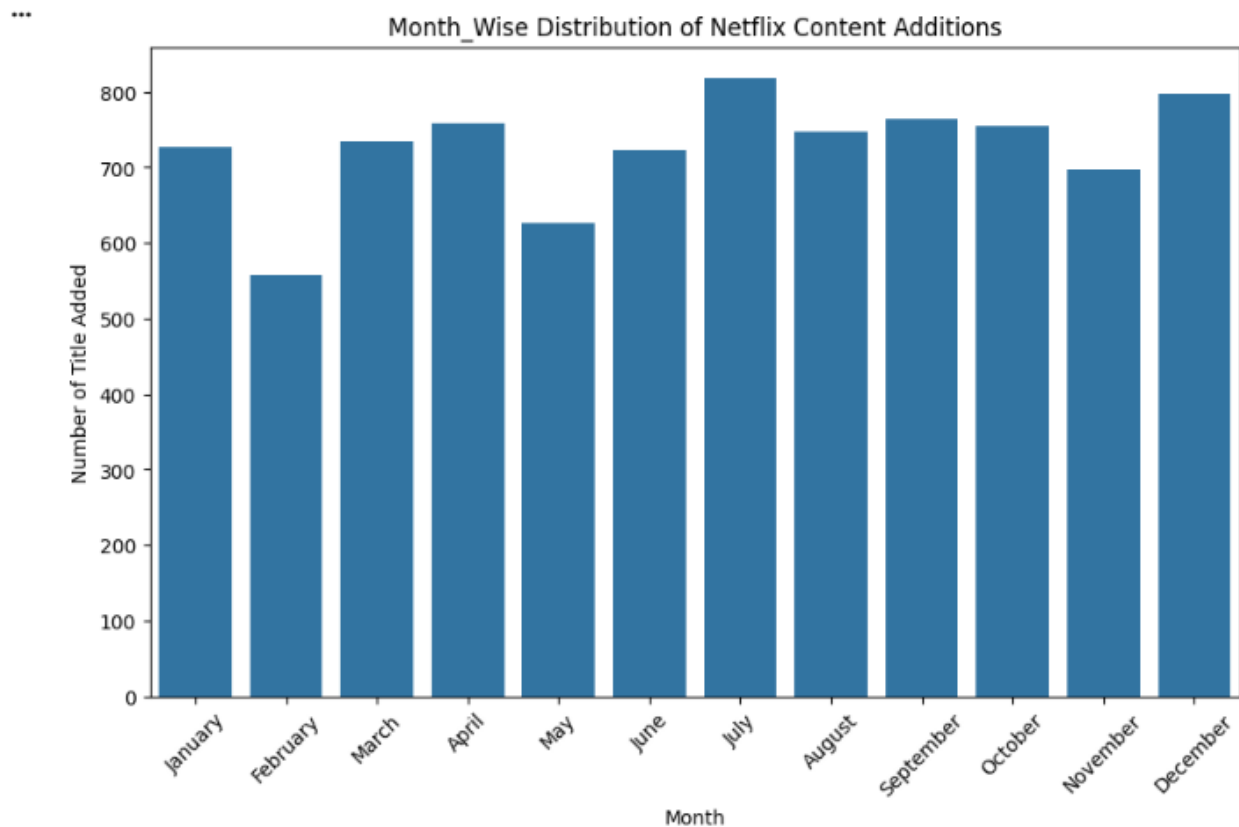
```
Month_trend["Month_name"] = Month_trend["Month"].map(Month_map)  
Month_trend
```

***	Month	Count	Month_name
0	1	727	January
1	2	557	February
2	3	734	March
3	4	758	April
4	5	626	May
5	6	724	June
6	7	819	July
7	8	748	August
8	9	764	September
9	10	755	October
10	11	697	November
11	12	797	December

BUSINESS CASE: NETFLIX - DATA EXPLORATION AND VISUALISATION

Plot Month-Wise Trend:

```
plt.figure(figsize = (10,6))
sns.barplot(data = Month_trend, x = "Month_name", y = "Count")
plt.xticks(rotation = 45)
plt.xlabel("Month")
plt.ylabel("Number of Title Added")
plt.title("Month_Wise Distribution of Netflix Content Additions")
plt.show()
```



Insights

- Content additions on Netflix are not uniformly distributed across months.
- **July** records the **highest number of content additions**, making in the peak month for releases.
- **December** and **April-September** also show **consistently high activity**, indicating strong mid-year and year-end release strategies.
- **February** has the **lowest number of additions**, suggesting slower content rollout early in the year.

BUSINESS CASE: NETFLIX - DATA EXPLORATION AND VISUALISATION

- Overall, Netflix appears to **cluster major releases in the second half of the year**, rather than spreading them evenly.

Business Implication:

- **Strategic Release Timing:**

Netflix likely aligns major content drops with **periods of higher user engagement**, such as mid-year vacations and year-end holidays.

- **Marketing Optimization:**

Increased marketing budgets and promotional campaigns should be **concentrated around high release months (July-December)** to maximize visibility and user acquisition.

- **Operational Planning:**

Lower activity months like **February** can be used for:

- Platform improvements.
- A/B testing recommendations.
- Preparing content pipelines for peak seasons.

- **Subscriber Retention Strategy:**

Consistent releases during peak months help reduced churn by continuously offering fresh content when user activity is naturally higher.

BUSINESS CASE: NETFLIX - DATA EXPLORATION AND VISUALISATION

Month-Wise Movie vs TV Show Trend:

```
Month_type_trend = (data.groupby(["Month_added",  
"type"]).size().reset_index(name = "Count"))
```

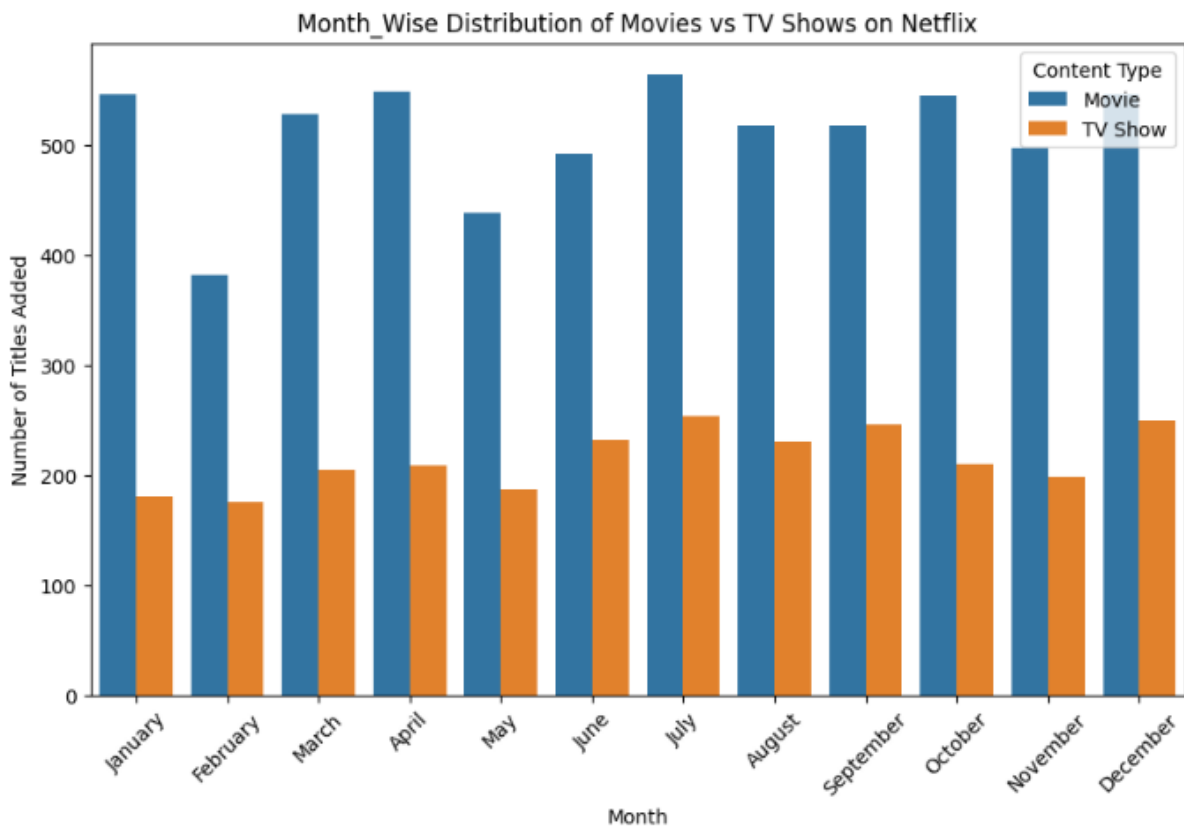
Month_type_trend

...	Month_added	type	Count
0	1	Movie	546
1	1	TV Show	181
2	2	Movie	382
3	2	TV Show	175
4	3	Movie	529
5	3	TV Show	205
6	4	Movie	549
7	4	TV Show	209
8	5	Movie	439
9	5	TV Show	187
10	6	Movie	492
11	6	TV Show	232
12	7	Movie	565
13	7	TV Show	254
14	8	Movie	518
15	8	TV Show	230
16	9	Movie	518
17	9	TV Show	246
18	10	Movie	545
19	10	TV Show	210
20	11	Movie	498
21	11	TV Show	199
22	12	Movie	547
23	12	TV Show	250

BUSINESS CASE: NETFLIX - DATA EXPLORATION AND VISUALISATION

Plot Month-wise distribution:

```
Month_type_trend["Month_name"] =  
Month_type_trend["Month_added"].map(Month_map)  
Month_order = ["January", "February", "March", "April", "May",  
"June", "July", "August", "September", "October", "November",  
"December"]  
plt.figure(figsize = (10,6))  
sns.barplot(data = Month_type_trend, x = "Month_name", y =  
"Count", hue = "type", order = Month_order)  
plt.xlabel("Month")  
plt.ylabel("Number of Titles Added")  
plt.title("Month_Wise Distribution of Movies vs TV Shows on  
Netflix")  
plt.xticks(rotation = 45)  
plt.legend(title = "Content Type")  
plt.show()
```



Insights

- Movies consistently outnumber Tv Shows across **all months**, indicating Netflix's stronger focus on movie additions throughout the year.

BUSINESS CASE: NETFLIX - DATA EXPLORATION AND VISUALISATION

- Content addition peak during the **mid-year to year-end period (July-December)** for both Movies and TV Shows.
- TV Shows additions show a relatively **steady and controlled increase** with higher counts observed in **July, September and December**.
- February and May show comparatively **lower content additions**, suggesting possible seasonal slowdowns or strategic release planning.

Business Implication:

- Netflix appears to follow a **content ramp-up strategy in the second half of the year**, likely to maximize user engagement during holidays and festive seasons.
- The dominance of Movies suggests the Netflix prioritizes **quick-consumption content**, which may help in attracting casual viewers and reducing content fatigue.
- Consistent TV Show releases across months help maintain **subscriber retention**, as episodic content encourages long-term platform engagement.
- Lower activity months (February, May) present an opportunity to **optimize release schedules** by introducing high-impact titles or marketing campaigns to balance user engagement throughout the year.

Best Time to Launch Tv Show:

```
Tv_data = data[data["type"] == "TV Show"]
```

```
Tv_data.head()
```

""	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description	Year_added	Month_added
1	s2	TV Show	Blood & Water	Unknown Director	Ama Oamata, Khosi Ngema, Gail Mababane, Thaban...	South Africa	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...	2021	9
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	Unknown Country	2021-09-24	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...	2021	9
3	s4	TV Show	Jailbirds New Orleans	Unknown Director	Unknown Actor	Unknown Country	2021-09-24	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...	2021	9
4	s5	TV Show	Kota Factory	Unknown Director	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train l...	2021	9
5	s6	TV Show	Midnight Mass	Mike Flanagan	Kale Siegel, Zach Gifford, Hamish Linklater, H...	Unknown Country	2021-09-24	2021	TV-MA	1 Season	TV Dramas, TV Horror, TV Mysteries	The arrival of a charismatic young priest brin...	2021	9

BUSINESS CASE: NETFLIX - DATA EXPLORATION AND VISUALISATION

```
Tv_month_trend =  
(Tv_data["Month_added"].value_counts().sort_index().reset_index()  
)  
Tv_month_trend.columns = ["Month", "Count"]
```

Tv_month_trend

***	Month	Count	
0	1	181	
1	2	175	
2	3	205	
3	4	209	
4	5	187	
5	6	232	
6	7	254	
7	8	230	
8	9	246	
9	10	210	
10	11	199	
11	12	250	

```
Tv_month_trend["Month_name"] =  
Tv_month_trend["Month"].map(Month_map)
```

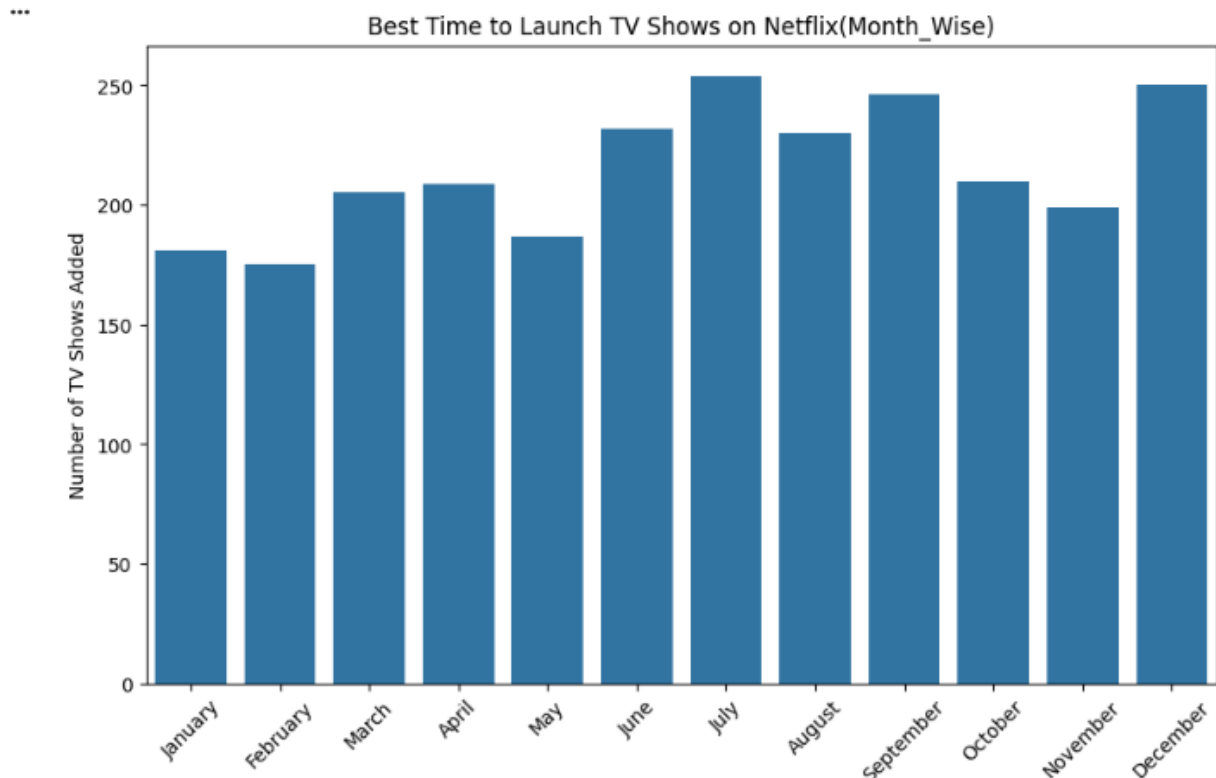
Tv_month_trend

***	Month	Count	Month_name	
0	1	181	January	
1	2	175	February	
2	3	205	March	
3	4	209	April	
4	5	187	May	
5	6	232	June	
6	7	254	July	
7	8	230	August	
8	9	246	September	
9	10	210	October	
10	11	199	November	
11	12	250	December	

BUSINESS CASE: NETFLIX - DATA EXPLORATION AND VISUALISATION

Plot best time to launch TV Shows:

```
plt.figure(figsize = (10,6))
sns.barplot(data = Tv_month_trend, x = "Month_name", y =
"Count")
plt.xticks(rotation = 45)
plt.xlabel("Month")
plt.ylabel("Number of TV Shows Added")
plt.title("Best Time to Launch TV Shows on Netflix(Month_Wise)")
plt.show()
```



Insights

- TV Shows additions on Netflix are **not evenly distributed across the year**.
- **July and December** show the highest number of TV Shows launches, indicating peak release activity.
- **June, September and August** also record relatively high numbers of TV show additions.
- **February and May** show comparatively lower launch volumes, suggesting off-peak periods.
- The overall pattern suggests that Netflix prefers launching TV shows during **mid-year and year-end periods**, likely aligning with user engagement and viewing time.

BUSINESS CASE: NETFLIX - DATA EXPLORATION AND VISUALISATION

Business Implication:

- **Optimal Launch Window:** Netflix should priorities **July and December** for launching high investment or flagship TV shows, as these months historically see higher release activity and likely higher audience attention.
- **Strategic Content Planning:** Mid-year (June-August) and year-end (December) periods can be leveraged for **season premieres, sequels and binge-worthy series**.
- **Marketing Alignment:** Marketing campaigns, promotions and subscriber acquisition efforts should be intensified around these peak months to maximize impact.
- **Off-Peak Opportunity:** Months with lower release (e.g. February and May) can be used to launch **niche or experimental content**, where competition for user attention is lower.
- **Release Calendar Optimization:** A data-driven release calendar can help Netflix balance content throughout the year while still capitalizing on high-engagement months.

Does Netflix focus more on TV Shows than Movies in recent years?

```
recent_data = data[data["Year_added"] >= 2015]
```

```
recent_data.head()
```

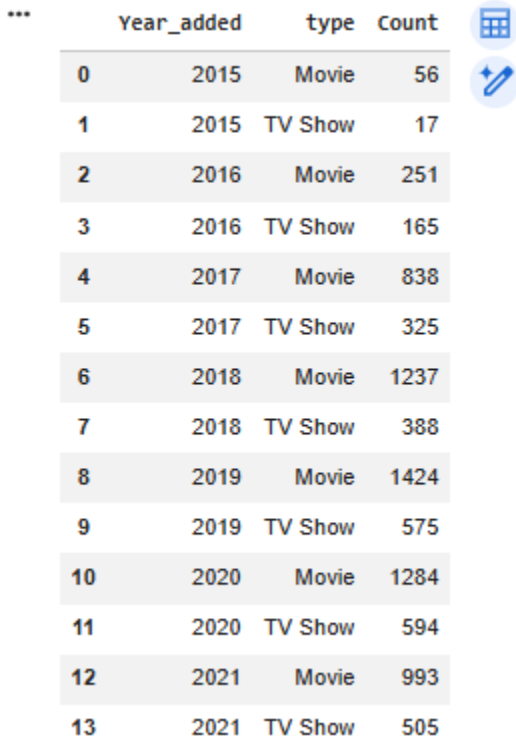
''	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description	Year_added	Month_added
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	Unknown Actor	United States	2021-09-25	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...	2021	9
1	s2	TV Show	Blood & Water	Unknown Director	Ama Oamata, Khosi Ngema, Gail Mababane, Thabani...	South Africa	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town l...	2021	9
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabil...	Unknown Country	2021-09-24	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lord...	2021	9
3	s4	TV Show	Jailbirds New Orleans	Unknown Director	Unknown Actor	Unknown Country	2021-09-24	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...	2021	9
4	s5	TV Show	Kota Factory	Unknown Director	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train l...	2021	9

To analyze Netflix's recent content strategy, the data from 2015 onwards was considered, as this period marks the beginning of Netflix's aggressive global expansion and original content production. This time window better represent Netflix's current strategic focus compared to the **initial years of limited content additions**.

BUSINESS CASE: NETFLIX - DATA EXPLORATION AND VISUALISATION

```
recent_trend = (recent_data.groupby(["Year_added",  
"type"]).size().reset_index(name="Count"))
```

```
recent_trend
```

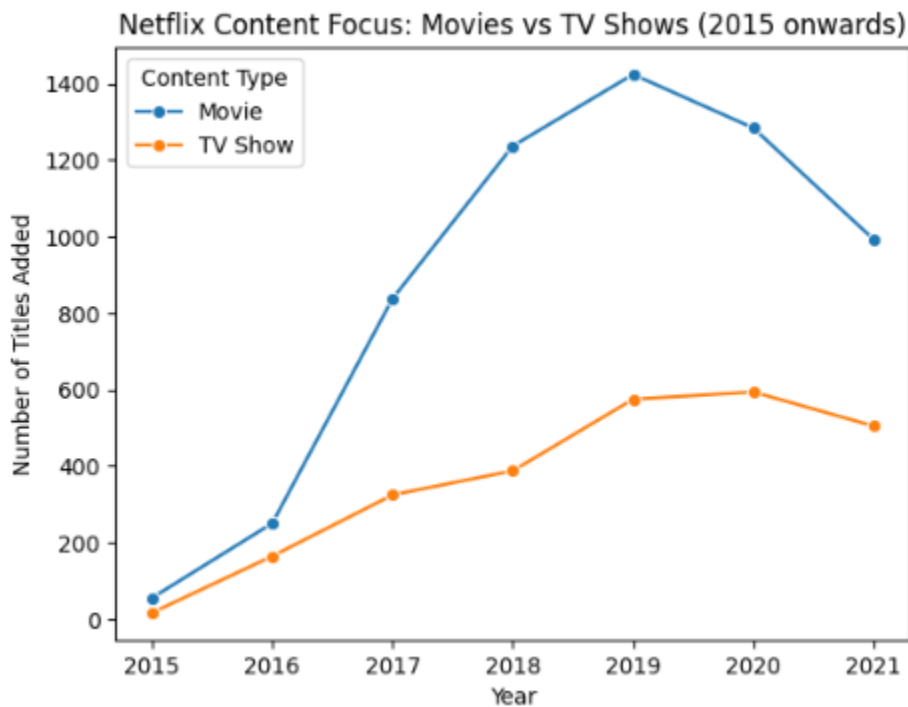


	Year_added	type	Count
0	2015	Movie	56
1	2015	TV Show	17
2	2016	Movie	251
3	2016	TV Show	165
4	2017	Movie	838
5	2017	TV Show	325
6	2018	Movie	1237
7	2018	TV Show	388
8	2019	Movie	1424
9	2019	TV Show	575
10	2020	Movie	1284
11	2020	TV Show	594
12	2021	Movie	993
13	2021	TV Show	505

Plot Netflix Content(2015 onwards):

```
sns.lineplot(data = recent_trend, x = "Year_added", y = "Count"  
, hue = "type", marker = "o")  
plt.title("Netflix Content Focus: Movies vs TV Shows (2015  
onwards)")  
plt.xlabel("Year")  
plt.ylabel("Number of Titles Added")  
plt.legend(title = "Content Type")  
plt.show()
```

BUSINESS CASE: NETFLIX - DATA EXPLORATION AND VISUALISATION



Insights

- From **2015 onwards**, Netflix consistently added more **movies** than **TV shows** every year with the gap widening significantly after **2017**
- A sharp increase in content additions is observed between **2016-2019** for both movies and TV shows, followed by a decline after **2020**, indicating a shift in content strategy.
- Although TV show additions increased over time, they never surpassed movies, suggesting TV shows are not Netflix's primary content type by volume.

Business Implication

- Netflix relies on movies to scale its library quickly and cater to diverse regional audiences due to faster production and broader appeal.
- Despite lower volumes, TV shows like play a key role in engagement and long term retention, supporting binge watching behavior.
- **The post-2020** decline suggests Netflix is focusing more on content efficiency, quality and ROI rather than aggressive volume expansion.

BUSINESS CASE: NETFLIX - DATA EXPLORATION AND VISUALISATION

Actor Analysis:

```
Actors_data = data.copy()
Actors_data["cast"] = Actors_data["cast"].str.split(", ")
Actors_data = Actors_data.explode("cast")

Top_actors = (Actors_data["cast"].value_counts().reset_index())
Top_actors.columns = ["Actors", "Count"]

Top_actors = Top_actors[Top_actors["Actors"] != "Unknown Actor"]
```

Top_actors

...	Actors	Count	
1	Anupam Kher	43	
2	Shah Rukh Khan	35	
3	Julie Teiwani	33	
4	Naseeruddin Shah	32	
5	Takahiro Sakurai	31	
...	...	...	
36137	Hina Khawaja Bayat	1	
36138	Yasa	1	
36139	Ali Al-Jabri	1	
36140	Mansoor Alfeeli	1	
36141	Michael Johnston	1	
36141 rows x 2 columns			

Top 10 Actors:

```
Top_10_actors = Top_actors.head(10)
```

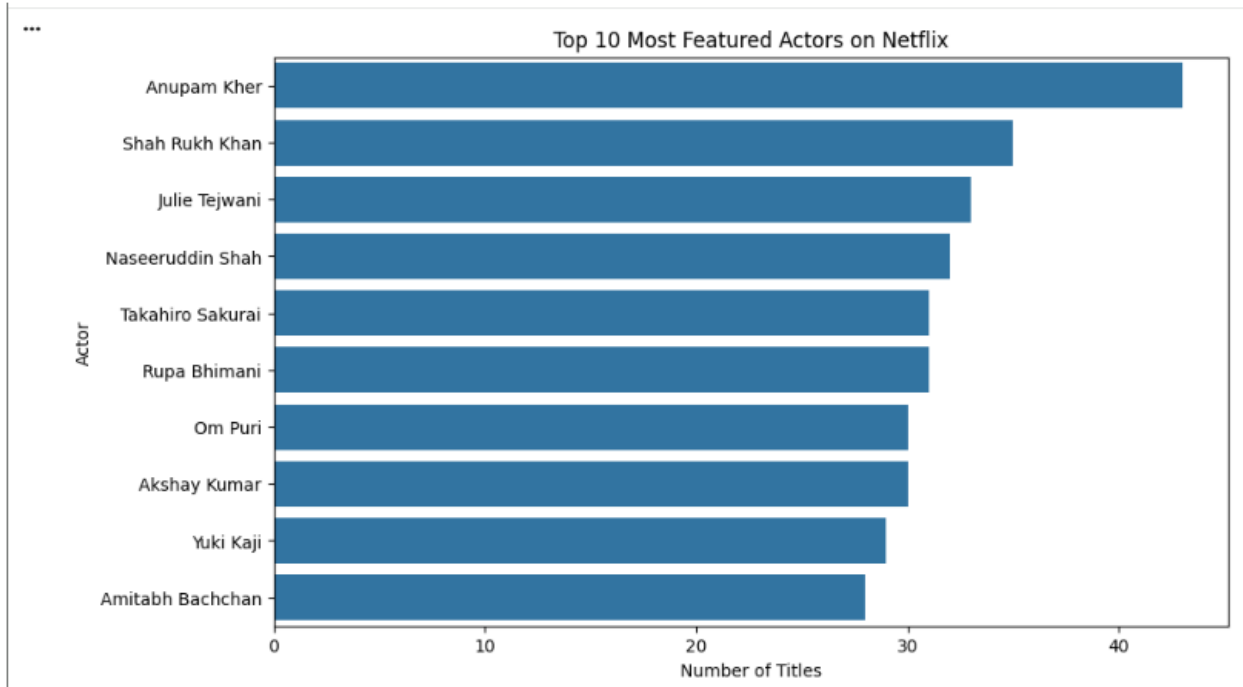
Top_10_actors

...	Actors	Count	
1	Anupam Kher	43	
2	Shah Rukh Khan	35	
3	Julie Teiwani	33	
4	Naseeruddin Shah	32	
5	Takahiro Sakurai	31	
6	Rupa Bhimani	31	
7	Om Puri	30	
8	Akshay Kumar	30	
9	Yuki Kaji	29	
10	Amitabh Bachchan	28	

BUSINESS CASE: NETFLIX - DATA EXPLORATION AND VISUALISATION

Plot Top 10 Actors on Netflix:

```
plt.figure(figsize = (10,6))
sns.barplot(data = Top_10_actors.head(10), x = "Count", y =
"Actors")
plt.title("Top 10 Most Featured Actors on Netflix")
plt.xlabel("Number of Titles")
plt.ylabel("Actor")
plt.show()
```



Insights

- The dataset contains a very large number of unique actors (**36141**) but most appear only once or twice. Only a small group of actors appear rapidly across multiple titles.
- Actors like **Anupam Kher (43)**, **Shah Rukh Khan (35)**, **Naseeruddin Shah (32)** etc. feature most frequently, indicating repeated collaborations or strong regional content pipelines (especially Indian Cinema).
- Many top actors belong to specific market like **India, Japan** reflecting Netflix's strategy of scaling regional content rather than relying only on **Global Hollywood Stars**.

Business Implication:

- Netflix can leverage frequently appearing actors as market anchor to attract and retain regional audiences especially in high-growth market in India.
- The long tail of actors suggest Netflix avoids over-dependence on a few stars, reducing cost risk and encouraging experimentation with new talent.

BUSINESS CASE: NETFLIX - DATA EXPLORATION AND VISUALISATION

- Netflix can further analyze performance metrics (ratings, completion rates) for top actors to guide future casting decisions and optimize ROI on star-led productions.

Director Analysis:

```
Directors_data = data.copy()
Directors_data["director"] =
Directors_data["director"].str.split(", ")
Directors_data = Actors_data.explode("director")
```

```
Top_directors =
(Directors_data["director"].value_counts().reset_index())
Top_directors.columns = ["Directors", "Count"]
```

```
Top_directors = Top_directors[Top_directors["Directors"] !=
"Unknown Director"]
```

Top_directors

...	Directors	Count
1	Martin Scorsese	139
2	Cathy Garcia-Molina	125
3	Steven Spielberg	121
4	Rajiv Chilaka	121
5	Youssef Chahine	104
...	...	...
4519	Justin Krook	1
4520	Dan Krauss	1
4521	Rod Blackhurst, Brian McGinn	1
4522	Paul Taublieb, Jon Freeman	1
4523	Peter Pardini	1

4523 rows × 2 columns

BUSINESS CASE: NETFLIX - DATA EXPLORATION AND VISUALISATION

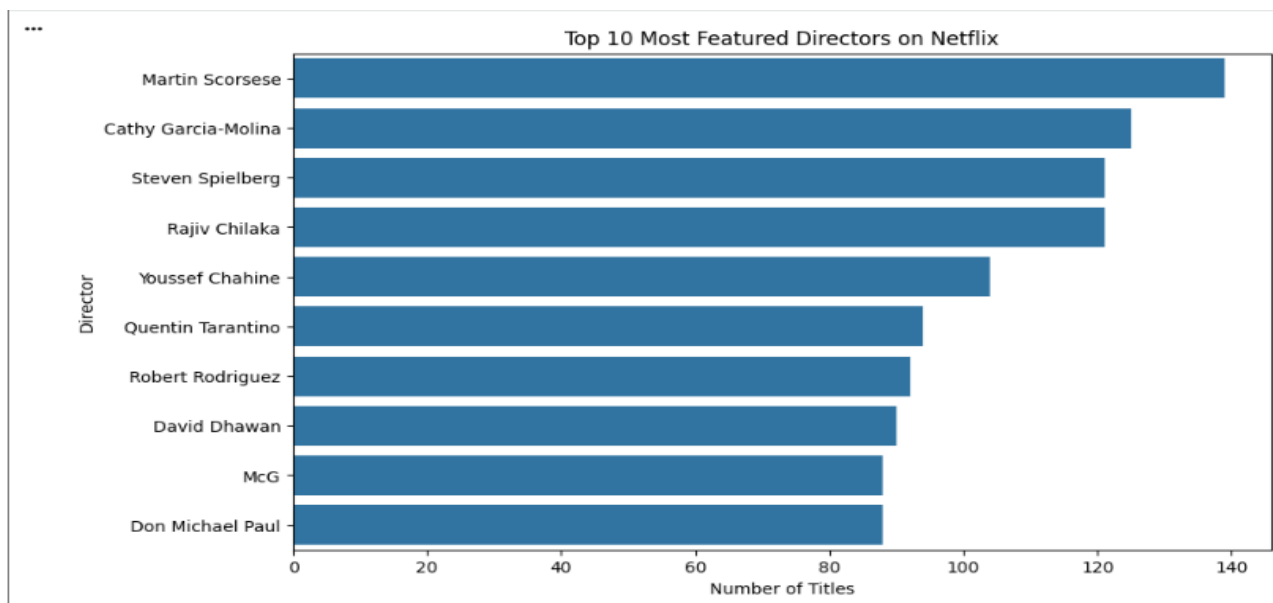
Top 10 Directors:

```
Top_directors.head(10)
```

...	Directors	Count
1	Martin Scorsese	139
2	Cathy Garcia-Molina	125
3	Steven Spielberg	121
4	Rajiv Chilaka	121
5	Youssef Chahine	104
6	Quentin Tarantino	94
7	Robert Rodriguez	92
8	David Dhawan	90
9	McG	88
10	Don Michael Paul	88

Plot Top 10 Directors on Netflix:

```
plt.figure(figsize = (10,6))
sns.barplot(data = Top_directors.head(10), x = "Count", y =
"Directors")
plt.title("Top 10 Most Featured Directors on Netflix")
plt.xlabel("Number of Titles")
plt.ylabel("Director")
plt.show()
```



BUSINESS CASE: NETFLIX - DATA EXPLORATION AND VISUALISATION

Insights

- A few directors account for a disproportionately large number of titles, while thousands of directors appear only once or twice.
- Director like **Martin Scorsese, Cathy Garcia-Molina, David Dhawan** etc. appear multiple times indicating trust in established high-performing creators.
- The top list includes directors from **Hollywood, Bollywood and other regions**, showing Netflix's multi-regional content strategy rather than a single-market focus.
- Most directors contribute only one title, highlighting experimentation and onboarding of new creators.

Business Implication:

- Netflix should continue long-term partnerships or exclusive deals with proven directors to reduce content risk and ensure consistent audience engagement.
- While established directors bring reliability, the long tail of new directors allows Netflix to discover fresh talent and control costs.
- Repeated appearances of *Indian and International directors* suggest opportunities to strengthen local languages originals using region successful creators.
- One-time directors can be tracked for performance, high-potential newcomers can be promoted into repeat collaborations.

Genre Analysis (Listed-in):

```
Genre_data = data.copy()
```

```
Genre_data["listed_in"] = Genre_data["listed_in"].str.split(",")
```

```
Genre_data = Genre_data.explode("listed_in")
```

```
Top_genres =
```

```
Genre_data["listed_in"].value_counts().reset_index()
```

```
Top_genres.columns = ["Genre", "Count"]
```

```
Top_genres
```

BUSINESS CASE: NETFLIX - DATA EXPLORATION AND VISUALISATION

**

	Genre	Count
0	International Movies	2752
1	Dramas	2427
2	Comedies	1674
3	International TV Shows	1328
4	Documentaries	869
5	Action & Adventure	859
6	Independent Movies	756
7	TV Dramas	739
8	Children & Family Movies	641
9	Romantic Movies	616
10	Thrillers	577
11	TV Comedies	550
12	Crime TV Shows	459
13	Kids' TV	433
14	Docuseries	380
15	Music & Musicals	375
16	Romantic TV Shows	364
17	Horror Movies	357
18	Stand-Up Comedy	343
19	Reality TV	244
20	Sci-Fi & Fantasy	243
21	British TV Shows	235
22	Sports Movies	219
23	Anime Series	172
24	Spanish-Language TV Shows	168



BUSINESS CASE: NETFLIX - DATA EXPLORATION AND VISUALISATION

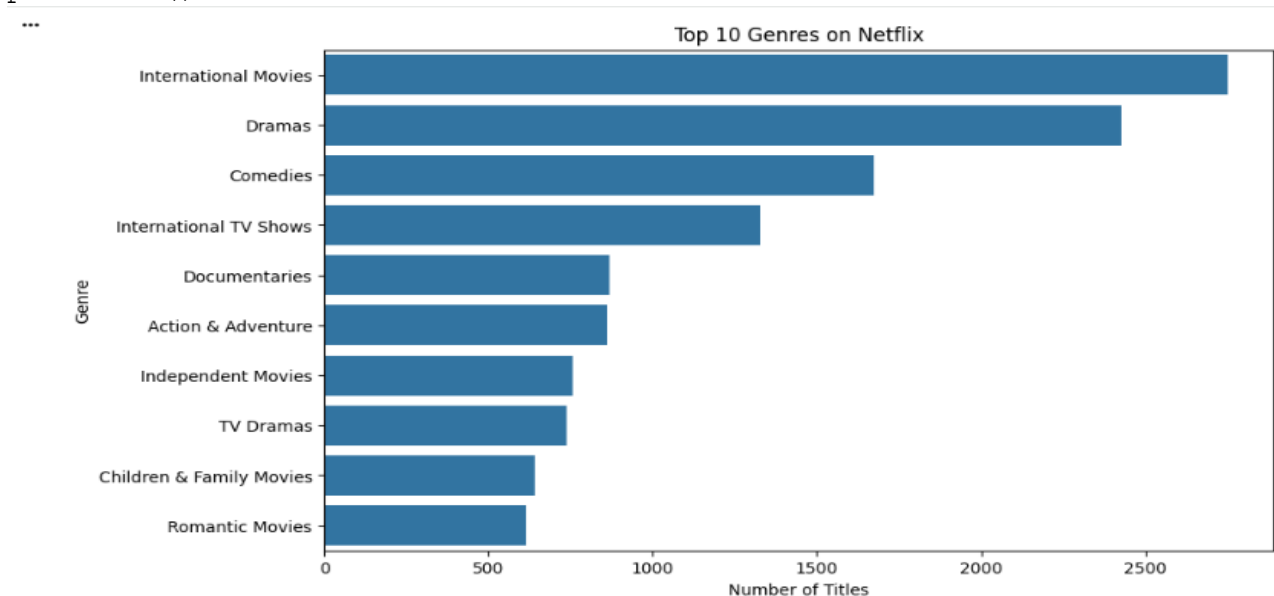
Top 10 Genre:

```
Top_genres.head(10)
```

	Genre	Count
0	International Movies	2752
1	Dramas	2427
2	Comedies	1674
3	International TV Shows	1328
4	Documentaries	869
5	Action & Adventure	859
6	Independent Movies	756
7	TV Dramas	739
8	Children & Family Movies	641
9	Romantic Movies	616

Plot Top 10 Genres on Netflix:

```
plt.figure(figsize = (10,6))
sns.barplot(data = Top_genres.head(10), x = "Count", y =
"Genre")
plt.title("Top 10 Genres on Netflix")
plt.xlabel("Number of Titles")
plt.ylabel("Genre")
plt.show()
```



BUSINESS CASE: NETFLIX - DATA EXPLORATION AND VISUALISATION

Insights

- *International Movies and International TV Shows* together form the largest share, showing Netflix's strong global content strategy beyond *Hollywood*.
- Dramas (both Movies and TV dramas) consistently rank at the top, indicating broad, cross-market appeal.
- **Comedies, Action and Adventure** are strong secondary Genre. These genres maintain high volume, suggesting they work well as engagement and retention drivers.
- Though smaller in count, **Documentaries and Independent Movies** form a meaningful niche. Their presence indicates Netflix caters to diverse and mature audience not just mass entertainment.

Business Implication:

- Netflix should continue investing in country specific content (local languages, culture), as international genres clearly scale well globally.
- Since drama performs strongly across formats, Netflix can prioritize dramas for flagships originals and long-term franchises.
- Comedies and Action genres are ideal for consistent drops to keep engagement high and reduce churn.
- Treat Documentaries and Indie Film as brand builder. These genres strengthen Netflix's brand image and attract niche, loyal audience.

Final Conclusion:

This case study analyzed Netflix's content dataset to understand patterns in content growth, format preference, release timing, regional distribution and creative collaborations. The analysis shows that Netflix's content expansion accelerated significantly after 2015, reflecting strategic shift toward aggressive global scaling and increased investment in original content.

Movies continue to dominate the platform in terms of volume, indicating Netflix's reliance on movies for frequent content refresh and broad audience appeal. However, TV shows demonstrate consistent and steady growth in recent years highlighting their increasing importance for long term user engagement and subscriber retention. The analysis of release timing suggests that mid-year and year-end months are optimal for launching TV shows, aligning with periods of higher content additions and viewer activity.

Genre analysis reveals a strong emphasis on International Movies, Dramas and Comedies confirming Netflix's global first strategy and focus on universally popular content categories. Actor and director analysis indicates that while a small group of established creators contribute repeatedly to the platform, a large number of creators appear only once allowing Netflix to balance reliability with experimentation and cost control.

BUSINESS CASE: NETFLIX - DATA EXPLORATION AND VISUALISATION

Overall, the findings suggest that Netflix's content strategy is driven by data-backed decisions that prioritize scale, international expansion and audience retention. By continuing to optimize its content mix, release time, genre focus and creator partnerships Netflix can further strengthen its competitive position and sustain long-term growth in the global streaming market.